

Nonparametric Predictive Regression with Exogenous Shocks: A Sieve Empirical Likelihood Approach (Version 2.0 AR1)

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1 Introduction and Data-Generating Process

We consider the problem of testing return predictability in the presence of highly persistent predictors and exogenous covariates (such as weather shocks). The data-generating process (DGP) is specified by the following system of equations:

1. Predictive Regression:

$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t \quad (1)$$

$$X_t = \theta + \rho X_{t-1} + \varepsilon_t \quad (2)$$

2. Endogeneity:

$$U_t = \phi \varepsilon_t + z_t \quad \text{with } E[z_t \mid \varepsilon_t] = 0 \quad (3)$$

where Y_t represents the asset returns, W_{t-1} is a stationary exogenous weather covariate, and X_{t-1} is a persistent financial predictor that is independent of W_{t-1} . The error terms are structured to accommodate potential endogeneity via the parameter ϕ .

Motivated by Perron (1989), our goal is to investigate return predictability while avoiding size distortions or false inferences caused by near unit roots in the predictor dynamics (i.e., $\rho \approx 1$). This is achieved by testing the null hypothesis of no predictability:

$$\begin{aligned} H_0 : \quad & \beta = 0 \\ H_1 : \quad & \beta \neq 0 \end{aligned} \quad (4)$$

By extending the empirical likelihood (EL) framework of Cai et al. (2026), we propose a two-stage Orthogonalized Sieve-EL framework that achieves a standard χ_1^2 limiting distribution under the null.

2 Motivating Empirical Example: El Niño and Cocoa Futures Returns

To illustrate the practical relevance of our proposed framework, we present an empirical scenario motivated by the agricultural commodity market shocks of 2023–2024. Let Y_t denote the excess returns on cocoa futures contracts, W_{t-1} represent an exogenous climatic indicator such as the Oceanic Niño Index (ONI) measuring El Niño–Southern Oscillation (ENSO) anomalies, and X_{t-1} denote a persistent financial predictor, such as the global commodity speculative net positions or the inventory-to-use ratio.

During the 2023–2024 period, a historically severe El Niño event induced extreme droughts in West Africa (primarily Côte d’Ivoire and Ghana, which account for nearly 60% of global cocoa supply). This led to a catastrophic supply deficit and triggered an unprecedented rally in cocoa futures, with prices tripling to exceed \$10,000 per metric ton. This setting provides a highly valid and clean empirical application of our model for several key reasons:

1. **Strict Exogeneity of Weather Shocks:** The climatic shock W_{t-1} is strictly exogenous to the financial markets. The dynamics of ENSO anomalies are governed by global atmospheric and oceanic circulation systems, meaning they are independent of futures trading volume, inventory levels, or financial speculative activity (X_{t-1}). Thus, W_{t-1} is strictly independent of X_s for all t and s , validating Assumption A.
2. **Nonlinear Climate-Yield Relationship:** The impact of temperature and rainfall anomalies on cocoa production is highly nonlinear. Mild deviations in ONI may have negligible effects due to soil moisture buffering, whereas extreme anomalies exceeding threshold levels lead to severe crop disease (e.g., black pod disease) and crop failure. A linear specification would fail to capture this threshold-like behavior, necessitating a nonparametric formulation $m(W_{t-1})$ that can be flexibly approximated using the sieve basis in our first step.
3. **High Persistence of the Financial Predictor:** Financial predictors like inventory-to-use ratios or speculative interest rates are highly persistent and are often modeled as near unit root processes (i.e., $\rho \approx 1$). Performing standard OLS inference on β in the presence of $m(W_{t-1})$ and a persistent X_{t-1} would generate severe size distortions.
4. **Separability and the Orthogonalized Sieve-EL Solution:** Our proposed framework allows the researcher to first filter out the complex, nonlinear weather impact $m(W_{t-1})$ using the sieve projection (Step 1). Because W_{t-1} and X_{t-1} are independent, this projection is consistent. By subsequently centering the financial predictor’s weights against the sieve basis, we purge the estimation error of the weather curve from the empirical likelihood moment condition. This enables valid, standard χ^2_1 -based inference on the predictive power β of the persistent financial variable.

3 The Orthogonalized Sieve-EL Framework (Version 2.0 AR1)

We handle the unknown nonparametric curve $m(W_{t-1})$ by approximating it using a sieve basis of dimension K (e.g., polynomials or splines). Let $\mathbf{P}_K(W_{t-1}) = (p_1(W_{t-1}), \dots, p_K(W_{t-1}))'$ be the $K \times 1$ vector of these basis functions.

3.1 The Step-by-Step Algorithm

The estimation and inference procedure is implemented sequentially via the following five steps:

Step 1: The First-Stage AR(1) (Purging the Predictor)

Exactly like the Cai & Wang (2014) framework, we run a first-order autoregression for the persistent predictor X_t to extract its daily shocks:

$$X_t = \hat{\theta} + \hat{\rho}X_{t-1} + \hat{\varepsilon}_t \quad (5)$$

We save the estimated residuals, $\hat{\varepsilon}_t$.

Step 2: The Joint Sieve-Cai-Wang Estimation (Purging the Weather Curve)

Because the return shock U_t is endogenous ($U_t = \phi\varepsilon_t + z_t$), we must control for the predictor innovations. We run a joint ordinary least squares (OLS) regression of the returns Y_t on our Sieve basis $\mathbf{P}_K(W_{t-1})$, the persistent predictor X_{t-1} , and our saved residuals $\hat{\varepsilon}_t$:

$$Y_t = \mathbf{P}_K(W_{t-1})'\mathbf{c} + \beta X_{t-1} + \phi\hat{\varepsilon}_t + \text{error}_t \quad (6)$$

This OLS regression yields our estimated Sieve coefficient vector, $\hat{\mathbf{c}}$. Because this is a joint regression:

- **No Omitted Variable Bias:** Since X_{t-1} is included as a regressor, $\hat{\mathbf{c}}$ does not inappropriately absorb the predictor's mean ($\beta_0\mu_X$).
- **No Stambaugh/Endogeneity Bias:** Since the innovation $\hat{\varepsilon}_t$ is included as a regressor, the OLS estimators are algebraically purged of the correlation between the return shock U_t and the predictor shock ε_t .

Step 3: Redefining the Adjusted Return

We now subtract only the purged weather curve to obtain our adjusted dependent variable, which strictly isolates the financial signal:

$$\tilde{Y}_t = Y_t - \mathbf{P}_K(W_{t-1})'\hat{\mathbf{c}} \quad (7)$$

Step 4: The Sieve Weight Projection

To ensure the EL numerator remains completely clean of the remaining Sieve estimation error, we run an OLS regression of the raw weights $w(X_{t-1})$ (e.g., $\tanh(X_{t-1}/10)$) on the exact same Sieve basis $\mathbf{P}_K(W_{t-1})$. Our centered weight, w_t^c , is defined strictly as the residuals of this regression:

$$w_t^c = w(X_{t-1}) - \mathbf{P}_K(W_{t-1})' \hat{\gamma} \quad (8)$$

where $\hat{\gamma}$ is the OLS coefficient vector of the weight projected onto the weather basis.

Step 5: The Empirical Likelihood Moment Condition

For testing the null hypothesis $H_0 : \beta = \beta_0$, the EL moment condition is constructed as:

$$Z_t(\beta_0) = \left(\tilde{Y}_t - \beta_0 X_{t-1} \right) w_t^c \quad (9)$$

Because $\hat{\mathbf{c}}$ was consistently estimated in Step 2 using the full joint regression, we have:

$$\tilde{Y}_t - \beta_0 X_{t-1} \approx m(W_{t-1}) + U_t - \mathbf{P}_K(W_{t-1})' \mathbf{c} \quad (10)$$

$$Z_t(\beta_0) \approx (U_t + r_t) w_t^c \quad (11)$$

This condition is mathematically clean of any omitted variable constant bias. The individual moment conditions are properly centered around $(U_t + r_t) w_t^c$. Consequently, the sample variance (the denominator of the EL statistic) remains completely uninflated:

$$\frac{1}{T} \sum_{t=1}^T Z_t^2(\beta_0) \xrightarrow{p} \sigma_U^2 \sigma_w^2 \quad (12)$$

This completes the algebraic derivation. By mirroring the CCHT first-stage AR(1) purging step, the resulting EL ratio statistic converges elegantly to a standard, unscaled χ_1^2 limiting distribution:

$$\ell_T(\beta_0) \xrightarrow{d} \chi_1^2 \quad (13)$$

3.2 Comparison with Cai et al. (2026)

Our framework differs from the two-stage approach of Cai et al. (2026) in three fundamental aspects:

- **Nuisance Dimensionality:** Cai et al. (2026) handle a fixed scalar intercept α , whereas our model contains an infinite-dimensional nuisance function $m(W_{t-1})$, which requires a sieve approximation of dimension $K \rightarrow \infty$.
- **Orthogonalization Space:** In Cai et al. (2026), the weight centering $w^c(x) = w(x) - \frac{1}{T} \sum w(X_s)$ represents a projection onto the constant subspace $\text{span}\{\mathbf{1}\}$. In contrast, our orthogonalized weight w_t^c is projected onto the K -dimensional sieve space $\text{span}\{\mathbf{P}_K(W_{t-1})\}$.

- **Purging and Exogeneity:** By jointly estimating the sieve basis alongside X_{t-1} and $\hat{\varepsilon}_t$, we ensure the nonparametric components are fully immune to both omitted variable bias and Stambaugh endogeneity bias, mirroring the core philosophy of CCHT.

4 The Growth Rate of the Sieve Dimension K

To establish the asymptotic validity of our test statistic, we must place bounds on the growth rate of the tuning parameter K relative to the sample size T .

4.1 Lower Bound (Bias Elimination)

Assume the true weather curve $m(W)$ has smoothness s (i.e., is s times continuously differentiable). Sieve theory implies that the approximation error r_t satisfies $\sup |r_t| = O_p(K^{-s})$. To ensure that the approximation bias does not distort the asymptotic distribution, we require $\sqrt{T} \sum_{t=1}^T r_t w_t^c = o_p(1)$, which translates to:

$$\sqrt{T} K^{-s} \rightarrow 0 \quad \implies \quad K \gg T^{\frac{1}{2s}} \quad (14)$$

4.2 Upper Bound (Variance Control and Leakage Mitigation)

Estimating K parameters in OLS introduces in-sample variance inflation and potential "future leakage" bias. Although the α -mixing property of the weather process W_t with exponential decay prevents leakage from accumulating, the pure variance penalty scales as $K/\sqrt{T}$. To prevent variance inflation from inflating the denominator of the EL ratio, we require:

$$\frac{K}{\sqrt{T}} \rightarrow 0 \quad \implies \quad K \ll T^{\frac{1}{2}} \quad (15)$$

4.3 The Golden Ratio

Combining these bounds yields the formal asymptotic growth condition:

$$T^{\frac{1}{2s}} \ll K \ll T^{\frac{1}{2}} \quad (16)$$

If we assume $m(W)$ is twice-differentiable ($s = 2$), the condition becomes $T^{1/4} \ll K \ll T^{1/2}$. In practice, setting $K \approx T^{1/3}$ satisfies these bounds, ensuring that both approximation bias and variance inflation vanish asymptotically.

5 The Formal Independence Assumption

To ensure the algebraic orthogonalization rigorously eliminates the nonparametric estimation bias, we must formally state the economic relationship between

the financial market predictor and the exogenous weather shock. We impose the following two independence conditions on W_{t-1} :

Assumption 5.1 (Strict Exogeneity to the Market Shock) *The weather variable W_{t-1} is strictly exogenous to the financial market innovation U_t . Mathematically:*

$$\mathbb{E}[U_t \mid \mathcal{F}_{t-1}, W_{t-1}, W_t, W_{t+1}, \dots] = 0. \quad (17)$$

Remark. This condition ensures that using the entire dataset to estimate the weather curve does not inadvertently absorb future stock market shocks. It shuts down the "future leakage" trap. Because climatic processes are governed by atmospheric physics, weather shocks are completely oblivious to tomorrow's stock market crashes.

Assumption 5.2 (Cross-Sectional Independence from the Predictor) *The exogenous weather process $\{W_t\}$ is strictly independent of the financial predictor innovation process $\{v_t\}$. Consequently, W_{t-1} and X_{t-1} are independent random variables, implying:*

$$\mathbb{E}[w(X_{t-1}) \mid W_{t-1}] = \mathbb{E}[w(X_{t-1})]. \quad (18)$$

Remark. This assumption fundamentally preserves the predictor's signal. Because W_{t-1} and X_{t-1} are independent, projecting the EL weight $w(X_{t-1})$ onto the weather basis $\mathbf{P}_K(W_{t-1})$ produces an R^2 of exactly zero asymptotically. The sieve projection structurally deletes the nonparametric estimation bias without extracting a single drop of the persistent signal from X_{t-1} .

6 Conclusion

When researchers use standard predictive regressions, they are often forced to assume a constant intercept to maintain mathematical tractability. Consequently, all other exogenous macro-environmental factors are relegated to the error term. While this preserves the null hypothesis, it artificially inflates the residual variance, mathematically obliterating the test's power to detect true financial predictability. Our Orthogonalized Sieve-EL estimator, augmented with the CCHT first-stage AR(1) purging step, allows researchers to surgically remove arbitrary, non-linear environmental noise without violating the stringent requirements of Empirical Likelihood or suffering from Stambaugh bias. By doing so, the framework fully recovers the lost signal of the persistent predictor, offering a powerful tool for complex, real-world scenarios such as commodity pricing.